

8 An Introduction to Metabolism

KEY CONCEPTS

- 8.1** An organism's metabolism transforms matter and energy *p. 144*
- 8.2** The free-energy change of a reaction tells us whether or not the reaction occurs spontaneously *p. 147*
- 8.3** ATP powers cellular work by coupling exergonic reactions to endergonic reactions *p. 150*
- 8.4** Enzymes speed up metabolic reactions by lowering energy barriers *p. 153*
- 8.5** Regulation of enzyme activity helps control metabolism *p. 159*

Study Tip

Make a table: Fill in the following table for each process you read about in this chapter, such as water spilling over a dam or a chemical reaction.

Process	Starting Materials; Relative Energy Level	Ending Materials; Relative Energy Level	Will this process occur spontaneously (without an input of energy)?
Water spilling over a dam	Water at the top of a dam; higher energy level	Water at the bottom of a dam; lower energy level	Yes
Hydrolysis of ATP (Figure 8.9)			

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 8
- Animation: Exergonic and Endergonic Reactions
- Figure 8.10 Walkthrough: How ATP Drives Chemical Work

For Instructors to Assign (in Item Library)

- Activity: Energy Transformations
- Tutorial: How Enzymes Function

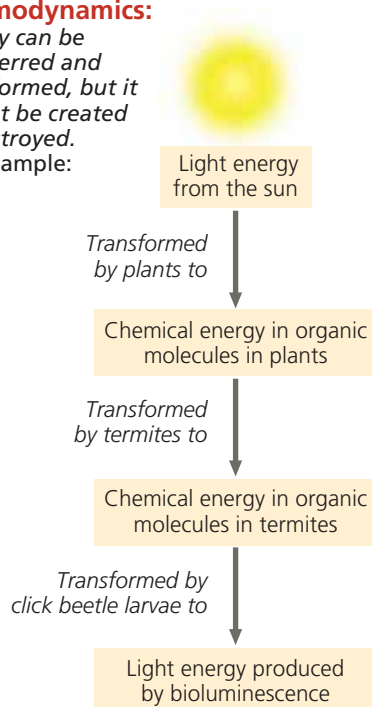


Figure 8.1 The green glowing spots on the outside of this Brazilian termite mound are larvae of the click beetle, *Pyrophorus nyctophanus*. These larvae convert the energy stored in organic molecules to light, a process called bioluminescence, which attracts termites that the larvae eat. Bioluminescence and other metabolic activities in a cell are energy transformations that are subject to physical laws.

How do the laws of thermodynamics relate to biological processes?

The first law of thermodynamics:

Energy can be transferred and transformed, but it cannot be created or destroyed. For example:



The second law of thermodynamics:

Every energy transfer or transformation increases the entropy (disorder) of the universe. For example, during every energy transformation, some energy is converted to thermal energy and released as heat:

